INCH-POUND

MIL-E-1/422G
AMENDMENT 3
10 August 1990
SUPERSEDING
AMENDMENT 2
26 January 1990

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 7YP1, 7YP2, AND 7YP7

This amendment forms a part of MIL-E-1/442G, dated 10 March 1976, and is approved for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
5	5
6	6

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Add the following under GENERAL:

* "Prior to shipment, 7YP2 tubes shall be electrically burned-in on a 100 percent basis. These tubes shall be operated at standard operating conditions with 5 microamperes direct current applied to each gun for a period of 48 hours. During this 48 hour period, the tube shall be turned off for 30 minutes (allowed to cool) and then turned on to continue burn-in at least 4 times. Following burn-in, each tube shall be tested in accordance with the following MIL-STD-1311 tests:

Method	Test
5201	Voltage Breakdown
5206	Gas "Cross"
5223	Modulation
5231	Spot displacement (leakage)
5241	Grid-Cutoff Voltage
5251	Grid No. 1 Leakage Current
5251	Anode No. 2 Leakage Current
5201	Electrode Current (Anode No. 1)

~~Those tubes which fail any of these tests shall not be shipped.~~

"Visual inspections shall be made to ensure that all JAN 7YP2 CRT's are failure free from arcing for continuous and intermittent operational testing during all manufacturing and acceptance testing."

* "Pressure test method 1141 shall be performed, except 30 pounds per square inch (PSI) absolute pressure shall be applied in lieu of 45 PSI absolute, on a 100 percent basis prior to shipment for the JAN 7YP2 tube only. Only those tubes meeting the acceptance criteria for this test shall be shipped.

* "When a P2 CRT screen is specified, the phosphors shall be in compliance with Electronic Industries Association TEPAC Publication, TEP 116-B, 'Optical Characteristics of Cathode Ray Tubes.'"

* Quality conformance inspection, part 1, method 5221, conditions column: Add "P2 screen for 7YP2 only."

* Quality conformance inspection, part 2, method 5221, notes column: Add "13". Method 5246, focusing voltage (zero-bias): Delete in its entirety. Method 5246, focusing voltage at cutoff: Delete title and substitute "Focusing voltage at Ib = 25 microamperes direct current"; add "450" to the minimum limits column.

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* Quality conformance inspection, part 2 (continued), method 1111, conditions column: Add "Sample size shall be 1 percent of lot (lot = 1 month's production.)" In maximum limits column, add:"0 failures". Method 1101, notes column: Add "15". Method 1105, method column: Delete " G ". In the requirements or test column, add: "For qualifications only."

* Quality conformance inspection, part 3, life test, conditions column: "Delete "5,000 hours (7YP2 only)" and substitute "500 hours (see note 14)." (Reverses change made by amendment 2.) Methods 5111, 5216, 5115, and 1216, requirements or test column, add: "For qualification only."

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* Note 11: Add "This test applies only when P1 CRT screen phosphors are specified."

* Following note 12, add:

"13. This test is not required unless specified in the purchase order or contract.

"14. The tube manufacturer (7YP2 only) shall provide the user with two years of service life warranty commencing with shipment from the tube manufacturer. The second year of this warranty shall be pro-rated. Other terms and agreements regarding this warranty (i.e., pro-rating, handling of defectives) shall be negotiated at the time of purchase. The appropriate applicable documentation shall be provided by the tube manufacturer and this documentation shall accompany the tube. The user will be responsible for completing and returning the documentation to the manufacturer.

"15. This test shall be performed on 100 percent basis and torque shall be 20 pounds-inch (7YP2 only). Immersion test for qualification only."

The margin of this amendment is marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air-Force 80
DLA - ES

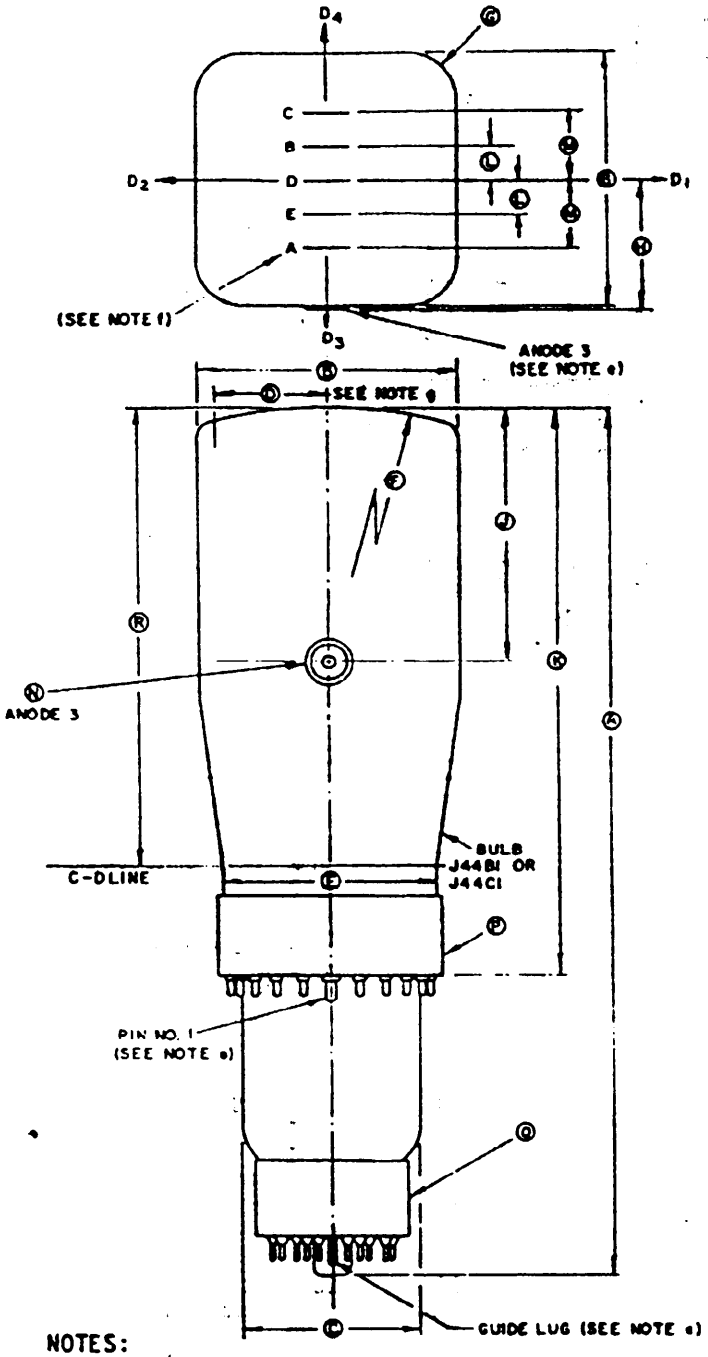
User activities:
Army - AV, ME
Navy - AS, CG, MC, OS
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3388)

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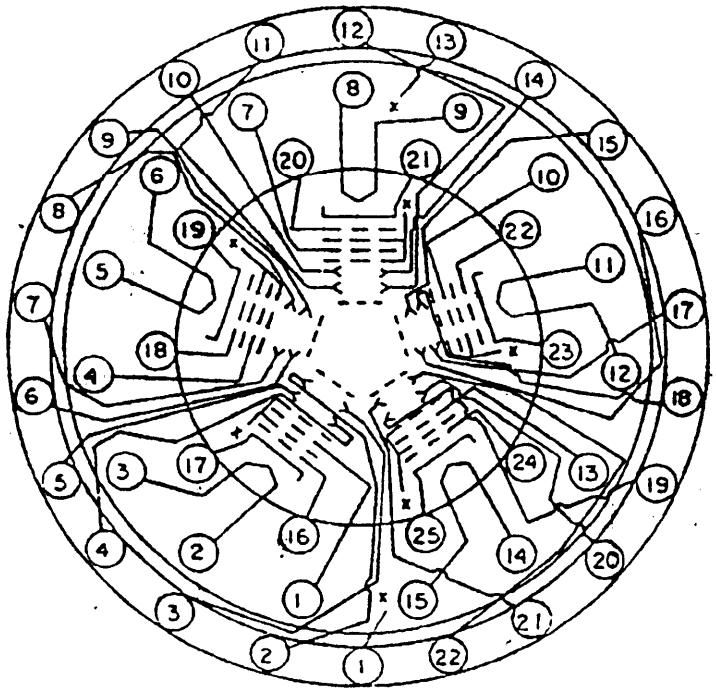
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	18.125 (460.38)	18.875 (479.42)
B	5.438 (138.13)	5.531 (140.49)
C	3.688 (93.68)	3.813 (96.85)
D	2.250 (57.15)	
E	4.344 (110.34)	4.468 (113.49)
H		2.797 (71.04)
J	5.125 (130.18)	5.375 (136.52)
K	12.125 (307.98)	12.375 (314.32)
Quality conformance inspection, part 3 (see note e)		
N	Bulb contact: J1-22	
P	Base, collar: A22-101	
S	Base, B25-139 (see note b)	
Reference dimensions (see note a)		
F	40.00 (1016.0)	
G	.75 (19.0)	
L	.75 (19.0)	
M	1.50 (38.1)	
R	10.00 (254.0)	

NOTES:
a. These dimensions are for information and are not required for inspection purposes.
b. For pin alignment use JEDEC gauge GB25-1.
c. Bulb contact J1-22, pin number 1 of base A22-101, and guide lug of base B25-139 shall each align with 3D4 trace within 10°.
d. Beam A shall center on position A as shown within .312 R. This requirement shall also apply to beams B, C, D, and E and their corresponding positions.
e. Useful screen radius.
f. Lancaster Glass bulb outline A-4314 or equivalent.

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2, and 7YP7.

Supersedes page 5 of amendment 2
of 26 January 1990.

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Bottom view

22 Pin collar base

Pin no.	Element	Beam
1	Accelerator	
2	Deflector D2	A
3	Deflector D1	A
4	Deflector D3	A
5	Deflector D4	A
6	Deflector D1	B
7	Deflector D3	B
8	Deflector D4	B
9	Deflector D2	B
10	Deflector D3	C
11	Deflector D1	C
12	Deflector D2	C
13	Accelerator	
14	Deflector D4	C
15	Deflector D1	D
16	Deflector D2	D
17	Deflector D4	D
18	Deflector D3	D
19	Deflector D4	E
20	Deflector D3	E
21	Deflector D2	E
22	Deflector D1	E

25 Pin base

Pin no.	Element	Beam
1	Focusing electrode	A
2	Heater	A
3	Heater	A
4	Focusing electrode	B
5	Heater	B
6	Heater	B
7	Focusing electrode	C
8	Heater	C
9	Heater	C
10	Focusing electrode	D
11	Heater	D
12	Heater	D
13	Focusing electrode	E
14	Heater	E
15	Heater	E
16	Grid number 1	A
17	Cathode	A
18	Grid number 1	B
19	Cathode	B
20	Grid number 1	C
21	Cathode	C
22	Grid number 1	D
23	Cathode	D
24	Gride number 1	E
25	Cathode	E

FIGURE 1. Outline drawing of electron tube types 7YP1, 7YP2, and 7YP7 - Continued.

Supersedes page 6 of amendment 2
of 26 January 1990.

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